

Register-Token Sidecar for Efficient Shadow-Mode Data Pipelines in End-to-End Autonomous Driving

[YOUR FULL NAME HERE]

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Theory & Validation Blueprint

Independent research proposal. Not affiliated with Tesla, Valeo, or the authors of DrivoR.

Abstract

End-to-end autonomous driving systems rely on massive fleets for continuous improvement through shadow-mode data collection. While current trigger heuristics effectively reduce upload volume compared to raw video, two open questions remain: (1) whether an orthogonal signal can surface long-tail failures that existing triggers miss, and (2) whether uploaded payloads can be further compressed while preserving downstream training signal.

We propose a lightweight register-token sidecar inspired by the DrivoR architecture [1]. A LoRA-tuned head with 16–32 camera-aware register tokens is run in inference-only mode alongside the main policy network. We formalize two testable hypotheses: register statistics as a hard-example mining signal (H1) and register tokens as a semantic compression codec (H2). We provide a complete 3-week offline validation plan, pseudocode, clear kill criteria, and risk assessment. No changes are made to the core pixels-to-controls policy network.

This sidecar is designed to be a low-risk, high-leverage data-pipeline upgrade compatible with any large-scale end-to-end fleet.

1. Introduction

Modern end-to-end autonomous driving programs, including fleet-scale systems such as Tesla FSD v12+ and similar stacks, have largely eliminated explicit maps and symbolic world models in favor of pure neural policies trained on vast fleet data. Shadow-mode collection — uploading only “interesting” clips triggered by disengagements, planner disagreement, or heuristics — is now standard practice to manage bandwidth and storage.

However, at planetary fleet scale, even highly optimized triggers leave room for improvement in two dimensions:

1. **Orthogonality of hard-example discovery** — Some long-tail failures remain under-sampled.

2. **Payload efficiency** — The current format (triggered imagery + dense features) is still expensive.

This whitepaper proposes a minimal, non-invasive sidecar that addresses both issues using the recently published DrivoR register-token primitive.

Concretely, the proposal contributes three things: a low-risk sidecar integration point, two testable hypotheses for data-pipeline improvement, and a short offline-first validation plan with explicit decision gates.

2. Background: DrivoR Register Tokens

DrivoR [1] introduces camera-aware register tokens into a pretrained DINOv2 ViT-S backbone. A small LoRA head (1–3 million trainable parameters) is added to produce 16–32 compact per-camera registers that replace the full set of patch tokens.

Key published results:

- More than 250x token compression
- No loss in trajectory prediction quality on NAVSIM v1/v2 and closed-loop HUGSIM
- Forward pass approximately 110 ms on a single A100 (single batch)

The registers act as a learned, compact scene descriptor while remaining fully compatible with downstream transformer decoders. We repurpose only the register mechanism as a passive sidecar for data efficiency.

3. Problem Statement

Current shadow-mode pipelines are effective but not exhaustive. We ask:

Question 1 (Orthogonality): Do register-token statistics provide a signal that correlates with real interventions above and beyond existing triggers?

Question 2 (Compression): Can register tokens + minimal triggered imagery serve as a drop-in replacement payload that preserves equivalent training signal?

4. Proposed Approach

We attach a frozen DINOv2 ViT-S + LoRA register head to the existing perception stack. The head runs in parallel in shadow mode only.

H1 — Hard-Example Mining Compute scalar statistics on the register tokens (assignment entropy, inter-register variance, distance to learned prototypes, etc.). High-score clips are hypothesized to be enriched for interventions missed by existing triggers.

H2 — Semantic Compression Transmit only the compact register tokens (a few KB) plus minimal triggered frames.

5. Experimental Validation Plan

Phase	Window	Primary Objective
Offline validation	Weeks 0–3	Test orthogonality and payload efficiency using public datasets or archived fleet clips.
Shadow-mode A/B	Weeks 4–7	Conditional deployment on a limited fleet if offline metrics justify progression.
Decision gate	End of Phase 2	Continue only if upload reduction, latency, and orthogonality targets are all met.

Phase 1 — Offline Validation (Weeks 0–3)

Use public datasets (nuScenes, BDD100K) or archived fleet clips.

- Run DrivoR register head on 1–10 million frames.
- H1: ROC-AUC of register-derived scores vs. intervention labels, conditioned on existing-trigger = negative.
- H2: Train lightweight downstream head on register-only vs. baseline payloads.
- Measure forward-pass latency on HW4-class hardware.

Phase 2 — Shadow-Mode A/B (Weeks 4–7, conditional)

Deploy inference-only sidecar on 50–100k vehicles.

Decision Gate (Kill Criteria)

Proceed only if all hold:

- At least 3x reduction in upload volume at parity intervention-discovery rate
- Added latency within Autopilot budget (less than 3 ms target on HW4)
- Statistically meaningful orthogonality (H1)

6. Illustrative Pseudocode

Listing 1: Inference-only register-token sidecar workflow.

```
1 vit = load_dinov2_s_backbone(frozen=True)
2 register_head = LoRA_RegisterHead(num_registers=16) # DrivoR weights
3
4 for clip in dataset:
5     features = vit(clip.frames)
6     registers = register_head(features)
7
8     entropy = compute_assignment_entropy(registers)
9     variance = compute_inter_register_variance(registers)
10    proto_dist = compute_distance_to_prototypes(registers)
11
12    score = combine_scores(entropy, variance, proto_dist)
13    log(score, clip.intervention_label, clip.existing_trigger_label)
```

7. Risks and Mitigations

- **Latency on HW4** — measured first.
- **Low orthogonality** — primary H1 kill condition.
- **Training-signal degradation (H2)** — verified with downstream head.
- **Scope creep** — strictly a data-pipeline sidecar.

8. Conclusion

The register-token sidecar offers a low-risk, high-leverage upgrade for any end-to-end autonomous driving fleet. Planned deliverables include open-source code and checkpoints upon completion of the offline phase.

We invite feedback, collaboration, or joint validation.

References

- [1] Kirby, E., et al. *Driving on Registers*. arXiv:2601.05083, Valeo.ai, January 2026. Presented at CVPR 2026. Available at: <https://arxiv.org/abs/2601.05083>.